MILITARY SPECIFICATION SHEET

FILTER ELEMENT, FLUID PRESSURE, HYDRAULIC, AIRCRAFT, DISPOSABLE, 5 MICRON ABSOLUTE

This specification sheet is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the filter element described herein shall consist of this document and the latest issue in effect of Specification MIL-F-83860.

REQUIREMENTS:

Bubble Point. The initial and burst out bubble point values shall meet the values established by the activity responsible for qualification and shall be determined as specified in 4.6.1 of general specification.

Filtration. The filter element shall remove 100 percent of all contaminants over 5 microns in size and 95 percent by weight of add contaminants.

Cold Start. The filter elements shall withstand 10 flow cycles as specified in 4.6.6 of general specification at -65°F.

Element Pressure Drop. The pressure drop through the clean element at 36gpm of MIL-H-5606 hydraulic fluid at 100 ± 10°F shall not exceed 11 psid.

Element Dirt Capacity. The filter element shall meet the dirt capacities specified in Table I, Contaminants.

Element Collapse. The filter element shall withstand a minimum differential pressures: Of 4500 psid

Description. The filter element shall be as shown in Figure 1 and specified herein.

FSC: 1650

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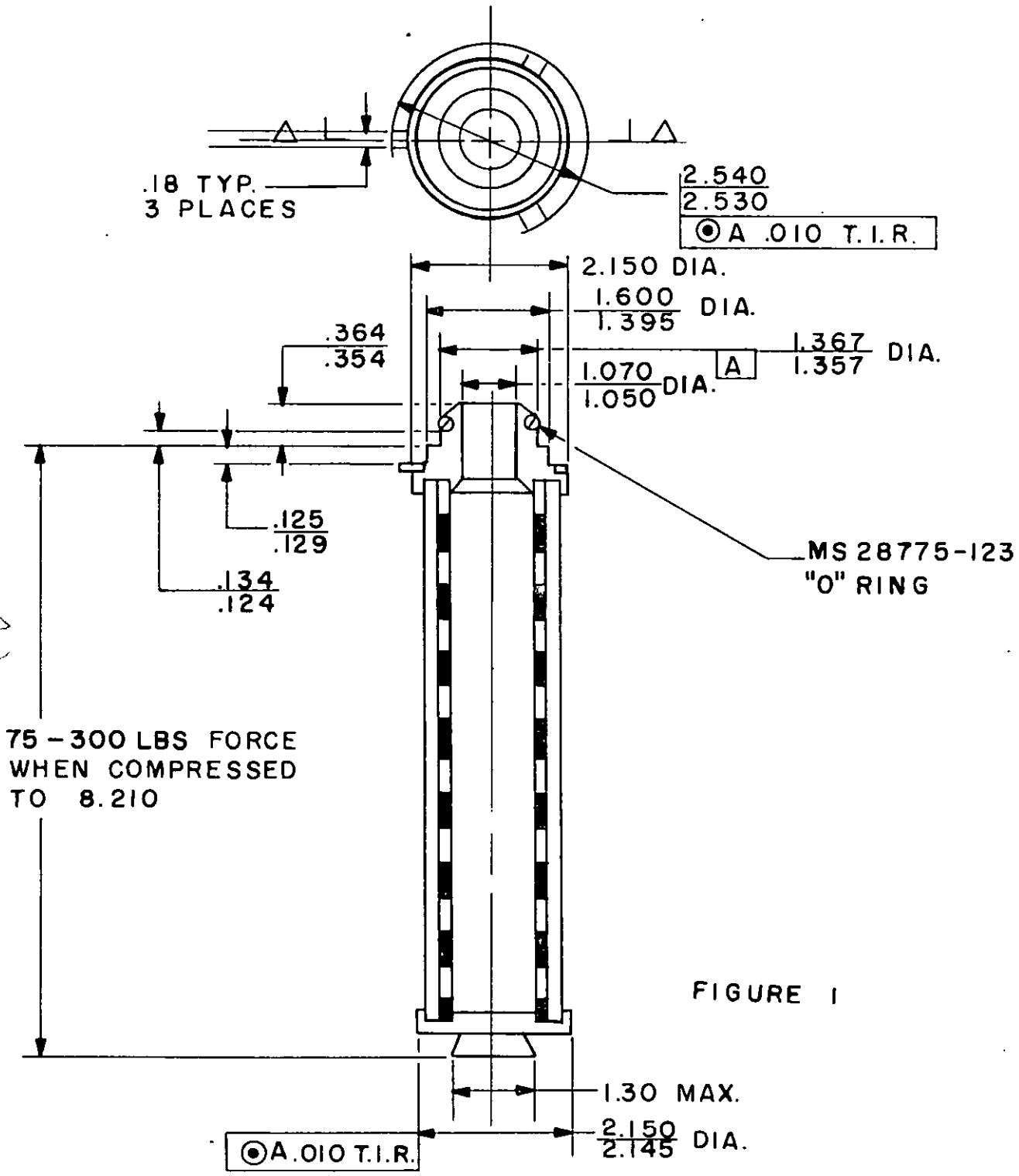


FIGURE 1

SECTION A-A

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Flow Fatigue. The filter element shall withstand the flow fatigue cycles specified at 300 cpm:

Cycles	Pressure Drop
1,120,000	20
112,000	50
12,500	100
400	1500
100	3000

Media Migration. There shall be no media migration traceable to the filter elements when tested in an appropriate housing, at vibration frequencies specified in 4.6.9 of general specification.

Maximum Particle Passed. The maximum particle passed shall be no greater than 5 microns.

TABLE I. CONTAMINANTS

	Degree of Filtration Test		Maximum Particle Test	Pressure Buildup Test	
Flow Rate (gpm)	Add Rate F-9 Beads (gms)	Add Rate AC Dust (gms)	Add Rate Carbonyl Iron E (gms)	Add Rate AC Dust (gms)	Minimum AC Dust to 90 psid (gms)
36.0	0.15	0.15	0.30	0.7	10.5

Part Number. The part number consists of an "M" plus the basic number of this specification sheet and a dash number.

Example: M83860/3

Custodian:
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